Thermodynamic foundations of the moist-parcel entropy in `tcpyPI`

A first-principles derivation of `entropy_S`, its conservation,
and the temperature inversion on the reversible adiabat

Companion to *An artificial discontinuity in the tcpyPI pressure solver*

July 22, 2026

Abstract

The potential-intensity solver in `tcpyPI` lifts an air parcel along a *reversible moist adiabat* and, at each pressure level, recovers the parcel temperature by holding one quantity fixed: the total specific entropy computed by `utilities.entropy_S`. This note derives that quantity, and everything around it, from first principles. We start from the statistical definition of entropy (Boltzmann/Gibbs), pass through the Sackur–Tetrode equation to the ideal-gas form $s = c_p \ln T - R \ln p + \text{const}$, treat the parcel as a multicomponent Dalton mixture with a condensed phase, derive the Clausius–Clapeyron relation and the latent-heat entropy from equality of chemical potentials, and assemble Emanuel's total specific entropy (E94, eq. 4.5.9)—the exact expression in the code. Three modern interludes deepen the statistical part: *canonical typicality*, which derives the Boltzmann weights of a small subsystem from a single pure state of one closed system—worked exactly on n harmonic oscillators, exceptional states included; a *flat-prior audit*, which demotes the equal-*a-priori* postulate to a gauge choice within an equivalence class of priors whose only invariant is a tilt criterion against the constraint's clamping slope; and the *Laplace–Legendre bridge* between the microcanonical $S(E)$ and the canonical $Z(\beta)$, including why the hard energy constraint and the soft Boltzmann weight are Laplace duals whose equivalence is guarded by two separations of scale. We then prove the entropy is conserved on the ascent (adiabatic $\Rightarrow$ no entropy flux; reversible $\Rightarrow$ no entropy production), give the microscopic Liouville/adiabatic -theorem reading of that conservation, and finish with the moist adiabat and the well-posed *inversion* $s(T, p) = s_0 \mapsto T$ that the solver performs by Newton iteration. Every formula is tied back to the specific function and constant it corresponds to in the source, and every figure is generated from the formulas derived here by `scripts/foundations_figs.py`—the inversion figure by running the solver's own algebra, cross-checked against the source docstrings. Numerical constants are quoted as in `tcpyPI/constants.py`; SI throughout, pressures excepted (hPa).

Contents

1 Scope and the target formula

The object we are after is the total specific entropy per unit mass of *dry* air, as coded in `utilities.entropy_S(T,R,P)`:

$$s \;=\; (c_{pd} + r_t\, c_\ell)\,\ln T \;-\; R_d\,\ln(p-e) \;+\; \frac{L_v\, r_v}{T} \;-\; r_v\, R_v\,\ln H, \tag{1}$$

with T the temperature (K), p the total pressure, e the partial pressure of water vapour, $p - e = p_d$ the partial pressure of dry air, r_t the total-water mixing ratio, r_v the vapour mixing ratio, L_v the latent heat of vaporisation, $H = e/e_s$ the relative humidity (capped at 1), and c_{pd}, c_ℓ, R_d, R_v material constants (Table 1). The code evaluates (1) at the parcel's launch state, where the air is unsaturated so $r_v = r_t = R$; that is why a single argument `R` feeds all water terms. When the parcel later saturates, the same s is held fixed and T is solved for—the *inversion* of §9.

Our task is to show where (1) comes from, why it is the right thing to conserve, and why the inversion is well posed. We build it in five movements: statistical mechanics (§2), classical thermodynamic scaffolding (§3), the multicomponent parcel (§4), phase equilibrium and latent heat (§5), and the assembly (§6). Conservation (§7), the moist adiabat (§8) and the inversion (§9) follow.

Symbol	Meaning	Value (`constants.py`)
c_{pd}	specific heat of dry air at const. p	$1005.7\,\mathrm{J\,kg^{-1}\,K^{-1}}$
c_{pv}	specific heat of water vapour at const. p	$1870.0\,\mathrm{J\,kg^{-1}\,K^{-1}}$
c_ℓ	(modified) specific heat of liquid water	$2500.0\,\mathrm{J\,kg^{-1}\,K^{-1}}$
R_d	gas constant of dry air	$287.04\,\mathrm{J\,kg^{-1}\,K^{-1}}$
R_v	gas constant of water vapour	$461.5\,\mathrm{J\,kg^{-1}\,K^{-1}}$
$\varepsilon = R_d/R_v$	mass ratio of the two gases	0.6220
L_v^0	latent heat at 0°C	$2.501 \times 10^6\,\mathrm{J\,kg^{-1}}$
$c_{pv} - c_\ell$	Kirchhoff slope $\mathrm{d}L_v/\mathrm{d}T$	$-630\,\mathrm{J\,kg^{-1}\,K^{-1}}$

Table 1: Material constants, quoted from `tcpyPI/constants.py`. The "modified" $c_\ell = 2500$ (rather than 4190) is a tuning of the reference, discussed in Remark 5.

Remark 1 (notational hazards)**.** Several symbols do double duty across the literatures this note joins; we keep each field's standard usage and flag the collisions once, here. Ω is the microstate *count* throughout §2—except in the ensemble table of §2.1, where the grand potential $\Omega(T, V, \mu)$ regrettably shares the letter (only the count appears beyond that table). $\rho(E)$, with an energy argument, is the density of states (§2.11); ρ with a state subscript (ρ_S, ρ_{mc}) is a density matrix (§2.4); and the ρ in the density temperature T_ρ (§8) refers to mass density. ε is a single-particle energy in §2 and retires before the mass ratio $\varepsilon = R_d/R_v$ of Table 1 takes over from §4 on. μ is a chemical potential in §2.1 and the mean occupancy M/n in the oscillator example of §2.4. H is relative humidity, while $\hat{H}$ is a Hamiltonian and $\mathcal{H}$ a Hilbert space; likewise R is a specific gas constant except as the code argument `R` of `entropy_S` (a mixing ratio), as discussed at (1). Context and fonts disambiguate; the double-duty is the price of quoting each source in its own dialect.

2 The statistical origin of entropy

This is the longest section, and it moves in four sweeps. First the *derivation* (§2.1–2.3): what the symbols mean operator-theoretically, and the Boltzmann distribution obtained the classical way, from a reservoir and an equal-probability postulate. Second, an *audit* of that derivation's most metaphysical input (§2.4–2.5): canonical typicality removes the ensemble altogether, and a companion argument shows the flat prior is merely the convenient representative of an enormous equivalence class. Third, the *computation* (§2.6–2.10): from Z through Sackur–Tetrode to $s = c_p \ln T - R \ln p$ for every ideal gas in the parcel. Fourth, the *reflection* (§2.11): the exact Laplace–Legendre bridge between the microcanonical $S(E)$ and the canonical $Z(\beta)$, closing with the hard-vs-soft-constraint reading of everything that came before. The payoff, §2.12, feeds the rest of the note.

2.1 The operator picture underneath

Before the classical development, it is worth stating the exact quantum-statistical object that development approximates, because it removes several apparent ambiguities in the symbol "$\sum_i$" that a careful reader will otherwise trip on. Rigorously, energy is not a number but the Hamiltonian *operator* $\hat{H}$ acting on the N-particle Hilbert space $\mathcal{H}_N$, and the partition function is a trace,

$$Z = \mathrm{Tr}\, e^{-\beta \hat{H}}, \tag{2}$$

whose value is independent of the basis in which it is evaluated. The microstate energies $\{E_i\}$ are the eigenvalues of $\hat{H}$; the index i labels an orthonormal energy eigenbasis; and the classical sum $\sum_i e^{-\beta E_i}$ used throughout this section is exactly (2) written in that eigenbasis, where $\hat{H}$ is diagonal with entries E_i. Three things this clarifies at once:

- **The index i is discrete, and $\mathcal{H}_N, \hat{H}$ depend on (V, N).** Confinement to a finite volume discretises the spectrum—a particle in a box of side $V^{1/3}$ has $\varepsilon \propto (n_x^2 + n_y^2 + n_z^2)\, V^{-2/3}$ with $n \in \mathbb{Z}_+^3$—so i is a genuine countable label, not a continuum. The parameters V and N do *not* index microstates; they select which Hilbert space and Hamiltonian one is on. Fixing (V, N), as the canonical ensemble does, means tracing over the eigenbasis of one definite operator $\hat{H}_{V,N}$.

- **The classical sum is the $\hbar \to 0$ limit.** The trace (2) reduces to the phase-space integral $(N!\, h^{3N})^{-1} \int \mathrm{d}\Gamma\, e^{-\beta H_{\mathrm{cl}}}$ of §2.7, with relative corrections $O(\hbar^2)$. The factors h^{3N} (one quantum state per phase-space cell of volume h^{3N}) and $N!$ (indistinguishability, from the (anti)symmetrisation $\mathcal{H}_N \subset \mathcal{H}_1^{\otimes N}$) are *not* classical inputs—they are residues of the quantum count, which is why they enter (26)–(24) seemingly by fiat.

- **Everything is a spectral functional of $\hat{H}$.** $Z(\beta)$ is the Laplace transform of the density of states $\rho(E) = \operatorname{Tr} \delta(E - \hat{H})$, and $F, \langle E \rangle, S$ all follow from it by one derivative each—spelled out in §2.6,

$$F = -k_B T \ln Z, \qquad \langle E \rangle = -\frac{\partial \ln Z}{\partial \beta}, \qquad S = \frac{\langle E \rangle - F}{T} = -\frac{\partial F}{\partial T}$$

  —so thermodynamics is the spectral geometry of the Hamiltonian. For N free identical particles the canonical Z_N is, exactly, a symmetric function of the single-particle Boltzmann weights $x_k = e^{-\beta \varepsilon_k}$ (one weight per single-particle level k), and *which* symmetric function records the exchange statistics:

  - the elementary $e_N(x) = \sum_{k_1 < \cdots < k_N} x_{k_1} \cdots x_{k_N}$ for **fermions** (strictly increasing indices $\Rightarrow$ no level used twice—Pauli exclusion);
  - the complete homogeneous $h_N(x) = \sum_{k_1 \le \cdots \le k_N} x_{k_1} \cdots x_{k_N}$ for **bosons** ($\le \Rightarrow$ any level may repeat, any occupation allowed);
  - the leading power-sum term $p_1(x)^N / N!$, with $p_1(x) = \sum_k x_k = z$, for the **Maxwell–Boltzmann count**—the classical $z^N/N!$ of (24) (derived in §2.6): particles taken as independent and then divided by $N!$ for indistinguishability, the dilute, non-degenerate limit (§2.7) in which the fermion/boson distinction washes out.

  So the indistinguishability that forces the $N!$ is the S_N-symmetry of these functions.

Finally, the various *ensembles* are Legendre transforms of one another, each holding fixed a different member of the conjugate pairs (E, T), (V, p), and (N, μ)—energy with temperature, volume with pressure p, and particle number with chemical potential μ (the free-energy cost of adding one particle); "summing over" a mechanical variable trades it for its conjugate:

sum runs over	held fixed	potential
microstates (E free)	E, V, N	$S(E, V, N)$ (microcanonical)
microstates	T, V, N	$F(T, V, N)$ (canonical)
microstates + N	T, V, μ	$\Omega(T, V, \mu)$ (grand)
microstates + V	T, p, N	$G(T, p, N)$ (isobaric)

(The grand potential $\Omega(T, V, \mu)$ shares its letter with the microstate count—Remark 1; only the count appears from here on.) The canonical development below fixes (T, V, N) and works with F; the parcel of the companion analysis ultimately wants (T, p)—the Gibbs potential G—which is why the specific entropy is finally expressed as $s(T, p)$ and inverted for T at prescribed p. What follows carries out (2) in its classical $\hbar \to 0$ form; the operator identity is the exact statement being approximated.

2.2 Boltzmann and Gibbs, and the system we will use

Entropy is, microscopically, a count of microstates. Fix the system once, for the whole section: a gas of N particles, each of mass m, confined to a container of volume V and held in equilibrium at temperature T. Its *macrostate* is specified by the three numbers (T, V, N); a *microstate* i is a complete specification of every particle's position and momentum (classically) or of the joint quantum state (quantally), and has a definite total energy E_i. This distinction—E_i is the energy of the *whole system* in microstate i, whereas ε (introduced in §2.7) will denote the energy of a *single particle*—is used consistently below.

For an *isolated* system all Ω microstates consistent with the macrostate are equally probable (the fundamental postulate of statistical mechanics), and Boltzmann's principle is

$$S \;=\; k_B \ln \Omega, \qquad k_B = 1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}}. \tag{3}$$

A parcel is *not* isolated—it exchanges energy with its surroundings—so its microstates are not equiprobable; they carry probabilities p_i, and the entropy is the more general Gibbs (information) form

$$S \;=\; -k_B \sum_i p_i \ln p_i, \tag{4}$$

which reduces to (3) when the p_i are uniform, $p_i = 1/\Omega$. To use (4) we must first find the p_i.

2.3 The Boltzmann distribution from energy conservation

The p_i are fixed by one physical input—*conservation of energy* between the system and its surroundings—plus the postulate above. Put the system in weak thermal contact with a large reservoir (subscript res throughout this subsection; not to be confused with the specific gas constant R that appears bare from §2.9 onward), and treat the composite (system + reservoir) as isolated with a fixed total energy

$$E_{\text{tot}} = E_i + E_{\text{res}} = \text{const.} \tag{5}$$

"Weak contact" means the interaction energy is negligible, so the energies are additive (5); "isolated composite" lets us apply the equal-*a priori*-probability postulate to it. When the system sits in one definite microstate i (energy E_i), the reservoir may be in any of its $\Omega_{\text{res}}(E_{\text{tot}} - E_i)$ microstates, all equally likely. Hence the probability of system-microstate i is proportional to that reservoir count,

$$p_i \;\propto\; \Omega_{\text{res}}(E_{\text{tot}} - E_i). \tag{6}$$

Because the reservoir is large, $E_i \ll E_{\text{tot}}$, so expand its *logarithm* (the extensive quantity) to first order:

$$\ln \Omega_{\text{res}}(E_{\text{tot}} - E_i) \;\simeq\; \ln \Omega_{\text{res}}(E_{\text{tot}}) - E_i \underbrace{\frac{\partial \ln \Omega_{\text{res}}}{\partial E}\Big|_{E_{\text{tot}}}}_{\equiv\, \beta}. \tag{7}$$

The coefficient β is a property of the reservoir alone; by (3) applied to the reservoir, $S_{\text{res}} = k_B \ln \Omega_{\text{res}}$, and the thermodynamic definition of temperature $1/T = \partial S_R/\partial E$, it is

$$\beta = \frac{1}{k_B}\frac{\partial S_{\text{res}}}{\partial E} = \frac{1}{k_B T}. \tag{8}$$

Exponentiating (7) and absorbing the E_i-independent factor into the normalisation gives the *Boltzmann distribution*

$$p_i = \frac{e^{-\beta E_i}}{Z}, \qquad Z = \sum_i e^{-\beta E_i}, \tag{9}$$

with the *partition function* Z enforcing $\sum_i p_i = 1$. The assumptions are now explicit: (i) equal *a priori* probabilities for the isolated composite; (ii) conservation of energy (5); (iii) a reservoir large enough that the linear term in (7) suffices; (iv) weak coupling, so energies add. (Equivalently, by Jaynes' maximum-entropy argument, (9) is the distribution that maximises (4) subject only to $\sum_i p_i = 1$ and a fixed mean energy $\langle E\rangle = \sum_i p_i E_i$, with β the Lagrange multiplier for energy—the two derivations agree.) Assumption (i) is the one that feels metaphysical—why should ignorance be *flat*?—and the next two subsections take it apart from opposite ends: §2.4 removes the ensemble altogether (a single pure state suffices), and §2.5 shows that even within ensemble language the flat prior is only the convenient representative of an enormous class of priors that all give the same physics.

2.4 The reservoir from the inside: canonical typicality and the n-oscillator example

The derivation just given works, but three of its four inputs are uncomfortable to a reader who believes the world is one closed system evolving unitarily. The reservoir is an external object introduced by hand; the equal-*a priori* -probability postulate (i) is imposed, not derived; and the ensemble language treats p_i as a subjective distribution over copies rather than a property of a single physical state. There is a modern reading—*canonical typicality* (Popescu, Short and Winter, *Nat. Phys.* **2**, 754 (2006); Goldstein, Lebowitz, Tumulka and Zanghì, *Phys. Rev. Lett.* **96**, 050403 (2006))—that produces the *same* Boltzmann weights from a single pure state of one closed system, with the postulate replaced by a theorem. It is worth a page, both because it dissolves the discomfort and because the oscillator makes every object in it explicit.

The general statement. Split the closed system into the piece we observe and the rest, $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_E$ ("S" the system, "E" its environment; dimensions d_S, d_E), and impose a global constraint—here a fixed total energy in a thin shell—by restricting to a subspace $\mathcal{H}_R \subseteq \mathcal{H}$ of dimension $d_R = \dim \mathcal{H}_R$. This d_R is exactly the microstate count Ω of (3), now read as a Hilbert-space dimension. The uniform (microcanonical) mixture on the shell is $\rho_{\rm mc} = \mathbb{1}_R/d_R$, and its reduction defines what PSW call the *canonical state* of the system,

$$\varrho_S \;\equiv\; \mathrm{Tr}_E\, \rho_{\rm mc} \;=\; \mathrm{Tr}_E \frac{\mathbb{1}_R}{d_R}. \tag{10}$$

Here Tr_E is the *partial* trace over the environment—the linear map from operators on $\mathcal{H}_S \otimes \mathcal{H}_E$ to operators on $\mathcal{H}_S$ fixed on product operators by

$$\mathrm{Tr}_E(A\otimes B) = A\, \mathrm{Tr}\, B \qquad \Big(\text{contrast the full trace } \mathrm{Tr}(A\otimes B) = \mathrm{Tr}\, A\, \mathrm{Tr}\, B\Big), \tag{11}$$

and on a general operator by linearity. It contracts the E factor to a number while leaving the S factor intact, so—unlike the full trace—it returns an *operator* on $\mathcal{H}_S$, not a scalar; (10) is thus a genuine density matrix on S, as is the pure-state reduction introduced next. Now take a *single* pure state $|\phi\rangle \in \mathcal{H}_R$—one definite vector, no ensemble—and form its reduced density matrix $\rho_S = \mathrm{Tr}_E\, |\phi\rangle\langle\phi|$. The theorem is that for the overwhelming majority of $|\phi\rangle$

(Haar measure[1] on the unit sphere of $\mathcal{H}_R$), ρ_S is indistinguishable from (10):

$$\langle\, \|\rho_S - \varrho_S\|_1 \,\rangle_\phi \;\le\; \sqrt{\frac{d_S}{d_E^{\mathrm{eff}}}}, \qquad d_E^{\mathrm{eff}} \equiv \frac{1}{\mathrm{Tr}(\varrho_E^2)}, \quad \varrho_E = \mathrm{Tr}_S\, \rho_{\mathrm{mc}}, \tag{12}$$

Two of the objects deserve unpacking before we use them. The *trace norm* $\|A\|_1 = \mathrm{Tr}\sqrt{A^\dagger A}$ is the sum of the singular values—for two states diagonal in a common basis, simply the ℓ^1 distance between the diagonals—and it is the operationally correct metric: the best single measurement distinguishes states ρ and σ with success probability $\frac{1}{2} + \frac{1}{4}\|\rho - \sigma\|_1$, so trace-norm-close means *physically* indistinguishable. And d_E^{eff}, the inverse purity of the environment's reduced microcanonical state, is its *participation number*—how many environment states are effectively occupied ($d_E^{\mathrm{eff}} = d_E$ exactly when all are weighted equally). By Lévy's lemma[2] (concentration of measure on the high-dimensional sphere), the fraction of states violating (12) by any fixed margin is exponentially small in d_R (nonempty but exponentially rare—the exceptional states, and the SHO's starkest one, are pinned down in Remark 2). The equal-*a priori* postulate is thereby *demoted to a theorem*: one need not assume a flat distribution over microstates—almost every individual pure state already *looks* flat to a small subsystem, because that is what generic entanglement across a large bipartition does. The weights p_i become eigenvalues of a reduced density matrix (an objective property of one state), not a subjective ensemble.

The n-oscillator realisation. Make $\mathcal{H}_S, \mathcal{H}_E, \mathcal{H}_R$ concrete with the harmonic oscillator, the one closed system where the counting is elementary. Take n identical oscillators

$$\hat{H} = \hbar\omega \sum_{i=1}^{n} \hat{n}_i \qquad \text{(zero-point dropped)}, \tag{13}$$

equivalently one particle in an isotropic n-dimensional harmonic trap. Let S be oscillator 1 and E the remaining $n-1$; the global constraint is a fixed total number of quanta $M = \sum_i k_i$, i.e. total energy $E_{\mathrm{tot}} = \hbar\omega M$. The shell $\mathcal{H}_R = \mathrm{span}\{\, |k_1, \ldots, k_n\rangle : \sum_i k_i = M \,\}$ has the stars-and-bars[3] dimension

$$d_R = \Omega(M, n) = \binom{M+n-1}{n-1}, \tag{14}$$

the microcanonical count for this system. A generic normalised state in it is $|\phi\rangle = \sum_{\sum k_i = M} c_{k_1\cdots k_n}\, |k_1, \ldots, k_n\rangle$ with Haar-random coefficients.

The reduced density matrix is automatically diagonal. Writing an environment configuration as $\vec{K} = (k_2, \ldots, k_n)$ with $|\vec{K}| \equiv \sum_{i\ge 2} k_i$, the constraint forces $k_1 = M - |\vec{K}|$ for *every* term, so each $\vec{K}$ pairs with a single system level. Hence

$$\rho_S = \mathrm{Tr}_E\, |\phi\rangle\langle\phi| = \sum_{k=0}^{M} \Big(\underbrace{\textstyle\sum_{|\vec{K}|=M-k} |c_{k,\vec{K}}|^2}_{\rho_S(k)} \Big)\, |k\rangle\langle k|, \tag{15}$$

with *no* off-diagonal entries: two different system energies would need the same environment state to carry two different energies, which the global conservation law forbids. This is

[1] The Haar (uniform) measure is the unique probability measure on the unit sphere of $\mathcal{H}_R$ invariant under every unitary; operationally, draw the components of $|\phi\rangle$ in any basis as i.i.d. complex Gaussians and normalise. This is also how Fig. 1 samples it.

[2] For a λ-Lipschitz function f on the unit sphere in real dimension $2d_R$, $\Pr\{|f - \langle f\rangle| \ge \epsilon\} \le 2\exp(-c\, d_R\, \epsilon^2/\lambda^2)$ with c an absolute constant: in high dimension, any smooth function of a random point is nearly constant. Applied to $f(\phi) = \|\rho_S - \varrho_S\|_1$ it upgrades the *mean* bound (12) to an exponential tail.

[3] Distribute M indistinguishable quanta over n oscillators: write M stars and $n-1$ separating bars in a row; every arrangement is exactly one occupation pattern $(k_1, \ldots, k_n)$, and there are $\binom{M+n-1}{n-1}$ arrangements.

dephasing by a conservation law made concrete—the microscopic reason ρ_S is a classical distribution over energy levels rather than a coherent superposition, so that the $\{p_i\}$ of (9) are literally its diagonal.

Its average is the canonical state. For a Haar-random unit vector in d_R complex dimensions $\langle |c_{k,\vec{K}}|^2\rangle = 1/d_R$, and the number of environment configurations with $|\vec{K}| = M - k$ is $\Omega(M-k,\, n-1)$, so

$$\langle \rho_S(k)\rangle = \frac{\Omega(M-k,\, n-1)}{d_R} = \frac{\binom{M-k+n-2}{n-2}}{\binom{M+n-1}{n-1}} \equiv \varrho_S(k), \tag{16}$$

which is exactly (10) written out—the marginal of the microcanonical shell onto one oscillator.

And a single state hugs that average. The diagonal entry $\rho_S(k)$ is a sum of $N_k = \Omega(M-k, n-1)$ terms $|c|^2$, each $\sim 1/d_R$; for a random unit vector its relative fluctuation is $1/\sqrt{N_k}$, exponentially small in n. The PSW bound (12) makes this sharp. Here the reduced microcanonical environment state ϱ_E is uniform over the d_R environment configurations with $|\vec{K}| \le M$ (hockey-stick identity $\sum_{j=0}^{M} \Omega(j, n-1) = d_R$), so $\mathrm{Tr}(\varrho_E^2) = 1/d_R$ and $d_E^{\mathrm{eff}} = d_R$. With $d_S = M + 1$ accessible levels,

$$\langle \|\rho_S - \varrho_S\|_1\rangle \le \sqrt{\frac{M+1}{\binom{M+n-1}{n-1}}} \approx \sqrt{\mu n}\, e^{-n\sigma(\mu)/2}, \qquad \sigma(\mu) = (1+\mu)\ln(1+\mu) - \mu\ln\mu, \tag{17}$$

where $\mu = M/n$ is the mean occupancy and $\sigma(\mu)$ is—not coincidentally—the canonical entropy per oscillator in units of k_B. The exponent is n times a thermodynamic entropy: the measure of atypical states is e^{-S/k_B}. Numbers make the point brutal. For a modest bath of $n = 100$ oscillators at $\mu = 1$ ($\beta\hbar\omega = \ln 2$), $d_R = \binom{199}{99} \approx 4.5 \times 10^{58}$ and the bound is below 10^{-28}: a *single* random pure state of one hundred oscillators already presents oscillator 1 with a density matrix that is thermal to twenty-eight decimal places—no reservoir, no ensemble, no postulate, only the geometry of a high-dimensional sphere. Figure 1 shows the collapse directly: the exact distribution of $\|\rho_S - \varrho_S\|_1$ over Haar-random shell states (whose diagonal is Dirichlet-distributed with parameters $N_0, \ldots, N_M$, which is how the figure samples it exactly) hugs the PSW bound and falls exponentially in n.

Remark 2 (where canonical typicality fails)**.** The bound (12)/(17) governs the *typical* state; its exceptional set is nonempty, and the oscillator names it. Typicality fails precisely for the states with little $S{:}E$ entanglement: a product $|\psi_S\rangle \otimes |\psi_E\rangle$ has $\rho_S = |\psi_S\rangle\langle\psi_S|$ pure, as far from thermal as a state can be. The extreme case is a single shell Fock vector with every quantum in the system,

$$|\phi\rangle = |M, 0, \ldots, 0\rangle, \qquad \rho_S = |M\rangle\langle M|, \qquad \tfrac{1}{2}\|\rho_S - \varrho_S\|_1 = 1 - P(M) = 1 - \tfrac{1}{d_R}, \tag{18}$$

using $P(M) = \Omega(0, n-1)/d_R = 1/d_R$. The trace distance is ≈ 1, the largest two density matrices can have, whereas the typical value (17) is $\sim 10^{-28}$—this lone state overshoots the bound by some twenty-eight orders of magnitude. Yet it costs the theorem nothing: such product states form a submanifold far below the sphere's $2d_R - 1$ real dimensions and are met with probability $\sim e^{-cn}$. They are exactly the definite-per-oscillator-occupation states—the natural stationary states of the *integrable* oscillator—so this kinematic exception is the shadow of the dynamical non-thermalisation in the cautionary tale below, and the reason a chaotic (ETH-obeying) $\hat{H}$, whose eigenstates are *themselves* typical, has no such exceptions.

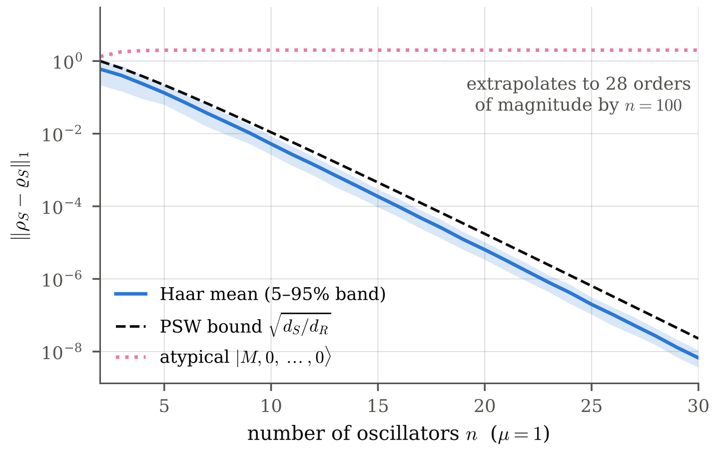

Figure 1: Canonical typicality, verified by exact sampling. For Haar-random states of the n-oscillator shell at $\mu = 1$, the trace distance $\|\rho_S - \varrho_S\|_1$ (mean and 5–95% band over 400 draws) tracks the PSW bound $\sqrt{d_S/d_E^{\text{eff}}}$ of (12) and decays exponentially in n—the slope is the entropy $\sigma(\mu)/2$ per oscillator of (17). The dotted line is the atypical product Fock state $|M, 0, \dots, 0\rangle$ of Remark 2, pinned at the maximal distance $2(1 - 1/d_R)$: the exceptional set exists but does not move.

Recovering the Boltzmann weight, and meeting §2.3. It remains to identify ϱ_S with the Gibbs state. This is the *equivalence of ensembles*, and for the oscillator it is one line: the consecutive ratio of (16) is

$$\frac{\varrho_S(k{+}1)}{\varrho_S(k)} = \frac{M-k}{M-k+n-2} \xrightarrow[M=\mu n]{n\to\infty} \frac{\mu}{\mu+1} \equiv x, \tag{19}$$

so $\varrho_S(k) \to (1-x)\,x^k$, a geometric (Planck) distribution (Fig. 2)—precisely $e^{-\beta\hbar\omega\,k}/Z_S$ with $Z_S = (1-x)^{-1}$ and $x = e^{-\beta\hbar\omega}$. The temperature is not an input: differentiating (14) gives $\beta = \partial \ln\Omega/\partial E = (\hbar\omega)^{-1} \ln\frac{M+n}{M} = -(\hbar\omega)^{-1}\ln x$, i.e. the very $\beta = \partial \ln \Omega_{\text{res}}/\partial E$ of (7)–(8), now read off the closed system's own microcanonical dial, and giving $\langle k\rangle = \mu = (e^{\beta\hbar\omega} - 1)^{-1}$, the Bose occupancy. The two logically distinct steps deserve separating: (12)/(17) (typicality, $\rho_S \to \varrho_S$) is the genuinely new, kinematic content that retires postulate (i); (19) ($\varrho_S \to$ Gibbs) is the old large-bath argument—and it *is* the Taylor truncation of §2.3, since $\ln\Omega(M{-}k, n{-}1) = (n{-}2)\ln(M{-}k) + \text{const}$ expands to $-\beta\hbar\omega\,k$ with a dropped quadratic term smaller by $O(1/n)$. The oscillator lets one *watch* both the postulate and the truncation of (7) become controlled, exact statements.

Why the oscillator is also the cautionary tale. Typicality is purely *kinematic*: (17) counts states and never used the dynamics. It guarantees thermal reductions are overwhelmingly abundant on the shell, but not that an *arbitrary* initial state *evolves into* them—that is the ergodicity question, and the bare oscillator fails it, instructively, for two structural reasons. First, within one shell every basis state has the same energy $\hbar\omega M$, so $e^{-i\hat{H}t/\hbar}$ acts as a single global phase and $\rho_S(t) = \rho_S(0)$ is frozen: the deliberately atypical start of Remark 2 (all M quanta in oscillator 1, $\rho_S = |M\rangle\langle M|$ pure) stays non-thermal forever. Second, across shells the spectrum is exactly equally spaced, so the whole system revives exactly every $2\pi/\omega$—there is no dephasing to carry an out-of-equilibrium state toward the typical set. Equivalently, the eigenstates are product Fock states $|k_1, \dots, k_n\rangle$, and a single one reduces to a *pure* $|k_1\rangle$, the opposite of thermal: the harmonic oscillator flatly violates the eigenstate-thermalisation property a chaotic $\hat{H}$ would enjoy. The cure is the one that also makes the spectrum generic—a

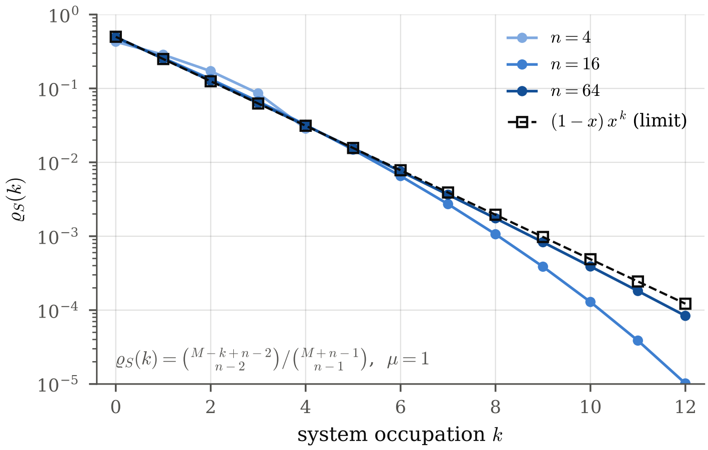

Figure 2: Equivalence of ensembles at finite n: the exact shell marginal $\varrho_S(k)$ of (16) at $\mu = 1$ against its $n \to \infty$ geometric (Planck) limit (19). The finite-n curves bend below the limit in the tail—that curvature is precisely the $O(1/n)$ quadratic term dropped in the Taylor truncation (7), and the support ends at $k = M$ because a finite bath cannot absorb more than every quantum. By $n = 64$ the first decade is indistinguishable from Planck.

weak *anharmonic* term $\frac{\alpha}{3}\sum_i(x_{i+1} - x_i)^3$ in the potential (the Fermi–Pasta–Ulam–Tsingou perturbation; note a *bilinear* coupling $\sum_i x_i x_{i+1}$ would not do—being quadratic it merely rotates to new normal modes and is exactly as integrable as before) lifts the degeneracy and the commensurability, entangles the eigenstates, and restores both relaxation and eigenstate thermality, at the cost of the closed-form counting above. Figure 3 shows both fates side by side—and adds the historical sting: the weakly anharmonic chain *shares* its energy yet very nearly *returns* it, the celebrated FPUT recurrence, so even where ergodicity is restored the approach to equipartition has long, model-dependent timescales. The division of labour is exactly the one worth recording: *typicality* (a theorem, and all the parcel's equilibrium thermodynamics needs) makes equilibrium states almost all of the shell, while *ergodicity* (a dynamical hypothesis the integrable oscillator lacks) is the stronger, separate input needed to *reach* them in time.

None of this changes a line of `entropy_S`: the $\beta = 1/k_BT$ and the Boltzmann weights are identical to §2.3. What changes is their standing—no external reservoir, no imposed postulate, one closed Hamiltonian system read through a single theorem. This is the same closed-system stance that §7 takes for the *conservation* of s on the ascent; §7.3 there and this subsection are its two faces—why the weights are canonical, and why they stay frozen—each dispensing with the reservoir in favour of one Hamiltonian.

2.5 The flat prior as a gauge choice: how little postulate (i) does

Typicality attacked postulate (i) by discarding the ensemble. This subsection attacks it from within ensemble language, and the conclusion is that the postulate is drastically overstated: what the physics actually requires of the prior is far weaker than flatness, and the flat choice is best understood as a *gauge fixing*—one convenient representative of an equivalence class of priors that are pairwise indistinguishable by any measurement in play.

The invariant. Strike postulate (i) and put *arbitrary* weights $w_i > 0$ on the shell microstates, $p_i = w_i / \sum_j w_j$. Rerunning the marginalisation of §2.3 for the energy ε of a small subsystem

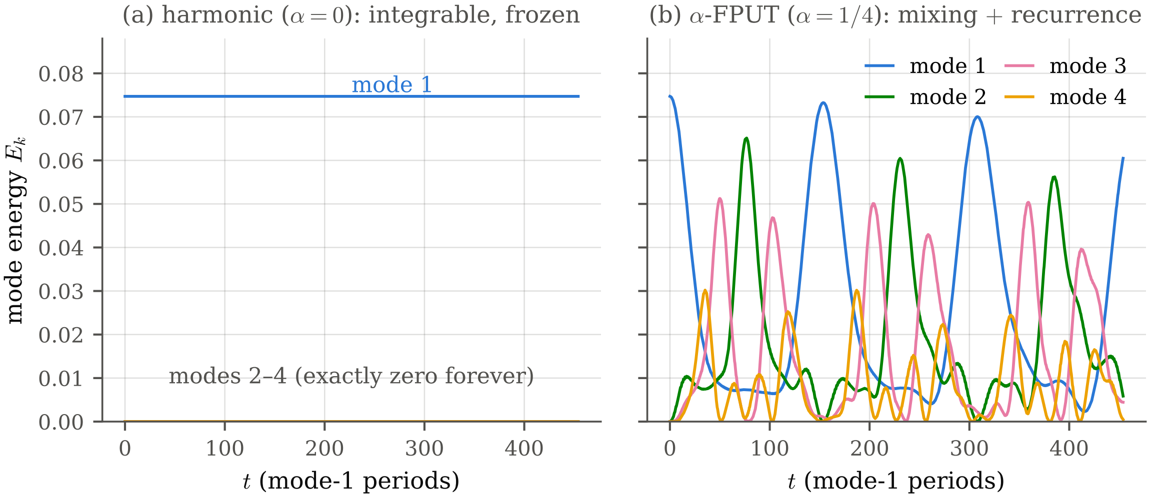

Figure 3: Ergodicity and its failure, integrated numerically for a 32-mass chain started with all its energy in normal mode 1. (a) The harmonic chain: every mode energy is a constant of motion, so the atypical initial state of Remark 2 persists forever—the integrable freeze. (b) The same chain with the weak cubic FPUT nonlinearity ($\alpha = 1/4$): energy now flows through modes 2–4, but instead of settling at equipartition it drains almost entirely back into mode 1 near $t \approx 155$ mode-1 periods and again near $t \approx 310$—the near-recurrences of the original 1955 experiment. Mixing is restored; the *timescale* to true equipartition remains long and model-dependent.

generalises (6) by one factor:

$$p(\varepsilon) \;\propto\; \bar{w}(\varepsilon)\,\Omega_S(\varepsilon)\,\Omega_B(E_{\text{tot}} - \varepsilon), \qquad \bar{w}(\varepsilon) \equiv \text{mean of } w_i \text{ over the bath states at that } \varepsilon \tag{20}$$

(with $w \equiv 1$ this is (43) of §2.11, where its Boltzmann limit is taken). The Boltzmann factor comes from $\ln \Omega_B$ falling at rate β per unit ε; the prior can disturb it only through $\bar{w}$. So the *entire* operational content of postulate (i) is the inequality

$$\boxed{\left|\frac{\mathrm{d}\ln \bar{w}}{\mathrm{d}\varepsilon}\right| \ll \beta = \frac{\mathrm{d}\ln \Omega_B}{\mathrm{d}\varepsilon}} \tag{21}$$

—the *coarse-grained* prior's tilt must be subdominant to the bath's clamping slope. Flatness is sufficient, memorable, and far stronger than necessary.

Wilder than expected: magnitude is free if uncorrelated. Notice what (21) does *not* constrain: the individual w_i. Each $\bar{w}(\varepsilon)$ averages over $\Omega_B = e^{S_B/k_B}$ bath states, so for weights fluctuating independently of ε (any distribution with finite variance), concentration gives $\bar{w}(\varepsilon) = \langle w \rangle (1 + O(e^{-S_B/2k_B}))$ for every ε at once: even *extensive* log-spread among the individual weights leaves the marginal Boltzmann to exponential accuracy, provided the wildness does not know about ε. And §2.4 has already demonstrated exactly this without saying so: a Haar-random state assigns weights $|c_i|^2$ that are i.i.d. exponential variables—relative fluctuation 100% per state, largest-to-smallest ratio of order $d_R \ln d_R$, i.e. log-spread of order S/k_B, about as far from flat as a prior can be—and Fig. 1 shows the resulting ρ_S thermal to 10^{-28}. *Canonical typicality is prior-robustness, quantified*: the theorem does not protect the flat prior; it proves the flat prior was never load-bearing.

The four tiers. The oscillator sorts every prior into a hierarchy, one line each:

1. **Tilt conjugate to the conserved total**, $w_i = e^{-\lambda M}$: constant on the shell (where M is fixed), hence *exactly* invisible—a pure gauge transformation. (Pleasingly, this is why β itself is physical only *across* shells.)

2. **Generic randomness of any magnitude**, w_i i.i.d.: washed out at $O(e^{-S_B/2k_B})$ by bath averaging, as above (Fig. 1). Gauge in practice.

3. **Smooth coarse tilt obeying** (21): shifts the saddle of §2.11 at subleading order; all thermodynamics unchanged in the limit.

4. **Tilt conjugate to a subsystem observable**, $w_i = e^{-\lambda k_1}$: the marginal ratio (19) acquires a factor $e^{-\lambda}$, i.e. $x \to xe^{-\lambda}$—oscillator 1 sits at a *different temperature*, $\beta_1\hbar\omega = \beta\hbar\omega + \lambda$. Visible for any $\lambda \sim O(1)$, because it competes with the clamping *slope*, not the clamping *range*. This is the genuinely wrong prior, and the atypical Fock state of Remark 2 is its $\lambda \to -\infty$ extreme: all weight on a single conspirator.

The restated postulate is then honest and minimal: *the prior must carry no tilt conjugate to an observable in play, beyond one negligible against β; everything else about it is unmeasurable.*

Whole-system quantities, dynamics, and Jaynes. For global thermodynamics the criterion is cruder still: any prior whose log-weights have subextensive spread has Shannon entropy $\ln\Omega + o(N)$, so S, $T = \partial S/\partial E$, and everything intensive are unchanged per particle —in large-deviation language, a prior with zero rate function cannot move a rate function, which is why ensemble equivalence never noticed the choice. Dynamics then explains which representative we write down: the flat (microcanonical) measure is the unique stationary one absolutely continuous with respect to Liouville measure for ergodic dynamics—the fixed point of the class—while the eigenstate-thermalisation property of §2.4 is the statement that chaotic evolution manufactures tier-2 genericity out of almost any initial weighting. Jaynes' maximum-entropy reading lands as a deliberate anti-climax: choose the flat prior not because nature is flat but because, within the class, it is the representative that adds no spurious information—and the class structure is precisely why the choice was never falsifiable. With the postulate audited from both ends, we can run the canonical machinery with a clear conscience.

2.6 From Z to entropy, and indistinguishable particles

Everything thermodynamic now follows from Z. Insert (9) into the Gibbs entropy (4), using $\ln p_i = -\beta E_i - \ln Z$:

$$S = -k_B \sum_i p_i\big(-\beta E_i - \ln Z\big) = k_B\beta \sum_i p_i E_i + k_B \ln Z = \frac{\langle E\rangle}{T} + k_B \ln Z, \tag{22}$$

where $\langle E\rangle = \sum_i p_i E_i = -\partial \ln Z/\partial\beta$ is the mean total energy (the thermodynamic internal energy). Defining the Helmholtz free energy $F \equiv -k_B T \ln Z$, equation (22) is

$$S = \frac{\langle E\rangle - F}{T} = -\Big(\frac{\partial F}{\partial T}\Big)_{V,N}, \tag{23}$$

the last equality being the standard thermodynamic identity (both V and N, the other natural variables of F, are held fixed). So the whole task reduces to computing Z for the parcel's constituents and differentiating.

Remark 3 (what depends on what)**.** It is worth being explicit about arguments, as the distinction drives the conservation argument later. The microstate energies E_i are *mechanical*: each is set by the Hamiltonian at a fixed volume and particle number, so $E_i = E_i(V, N)$—a function of the microstate label i and of (V, N), but *not* of temperature. Consequently the partition function and free energy carry the mechanical parameters as well, $Z = Z(\beta, V, N)$ and $F = F(T, V, N)$ —which is why (V, N) sit in the derivative subscript of (23) —while the *mean* energy $\langle E\rangle = \sum_i p_i E_i = U(T, V, N)$ is a thermodynamic function of state, belonging to the whole ensemble and to no single microstate.

A caution on the occupations, since it is easy to over-simplify: $p_i = e^{-\beta E_i}/Z$ is *not* a function of temperature alone. It depends on (T, V, N) jointly—on T through β and on (V, N) through E_i—so what fixing the macrostate determines is the *whole* distribution $\{p_i\}$, with no residual freedom. This is exactly the structure the reversible-adiabatic argument of §7 turns on. Under a slow (quasi-static) change of V the levels $E_i(V)$ shift, yet along an adiabat the product βE_i is held fixed for *every* i simultaneously—for a monatomic ideal gas $E_i \propto V^{-2/3}$ and the adiabat is $TV^{2/3} = \text{const}$, so $\beta E_i \propto T^{-1}V^{-2/3} = \text{const}$—and therefore each occupation p_i, and with it $S = -k_B \sum_i p_i \ln p_i$, is unchanged even as both T and the E_i move. "The levels shift; the occupations are frozen" is this statement, and it is what makes the ascent isentropic.

One structural fact makes Z tractable. For N *non-interacting* particles the total energy is a sum of single-particle energies, $E_i = \sum_{a=1}^{N} \varepsilon_a$, so the system sum factorises into a product of identical single-particle sums, z^N. But the particles are *indistinguishable*: permuting them yields the same physical microstate, so z^N overcounts by the $N!$ permutations. The corrected result—which resolves the Gibbs mixing paradox and is the classical limit of quantum statistics—is

$$Z = \frac{z^N}{N!}, \qquad z = \sum_{\text{single-particle states}} e^{-\beta\varepsilon}, \tag{24}$$

with z the single-particle partition function and ε a single-particle energy (contrast E_i, the whole-system energy of §2.2). Everything now hinges on z.

2.7 The translational partition function and the thermal de Broglie wavelength

For a structureless particle in a box of volume V the energy is purely kinetic, $\varepsilon = \mathbf{p}^2/2m$. In the classical (dilute, non-degenerate) regime the sum over single-particle states becomes a phase-space integral with one state per volume h^3 of phase space:

$$z_{\text{tr}} = \frac{1}{h^3}\int_V \mathrm{d}^3 r \int_{\mathbb{R}^3} \mathrm{d}^3 p\, e^{-\beta \mathbf{p}^2/2m} = \frac{V}{h^3}\prod_{i=x,y,z}\int_{-\infty}^{\infty} \mathrm{d}p_i\, e^{-\beta p_i^2/2m}. \tag{25}$$

Each of the three Gaussian integrals evaluates to $\int_{-\infty}^{\infty} e^{-\beta p^2/2m}\mathrm{d}p = \sqrt{2\pi m/\beta} = \sqrt{2\pi m k_B T}$, so

$$z_{\text{tr}} = \frac{V\,(2\pi m k_B T)^{3/2}}{h^3} \equiv \frac{V}{\Lambda^3}, \qquad \boxed{\Lambda \equiv \frac{h}{\sqrt{2\pi m k_B T}}} \tag{26}$$

The length Λ is the *thermal de Broglie wavelength*: the quantum coherence scale of a particle with thermal momentum $\sim \sqrt{m k_B T}$. It carries the entire temperature dependence of z_{tr}, as $\Lambda^3 \propto T^{-3/2}$. The classical treatment (24)–(25) is valid precisely when $n\Lambda^3 \ll 1$ (mean spacing $\gg$ coherence length); for air at surface conditions $n\Lambda^3 \sim 10^{-6}$, so the approximation is excellent.

2.8 Free energy and the Sackur–Tetrode entropy

Take a monatomic gas first, $z = z_{\text{tr}} = V/\Lambda^3$ (no internal structure). From (24), $\ln Z_N = N \ln z - \ln N!$, and with Stirling's approximation $\ln N! \simeq N \ln N - N$,

$$F = -k_B T \ln Z_N = -N k_B T\Big[\ln \frac{z}{N} + 1\Big] = -N k_B T\Big[\ln \frac{V}{N\Lambda^3} + 1\Big]. \tag{27}$$

The entropy follows from (23). Writing $\ln(V/N\Lambda^3) = \ln(V/N) + \frac{3}{2}\ln T + C_0$ with $C_0 = \frac{3}{2}\ln(2\pi m k_B/h^2)$ temperature-independent, only the $\frac{3}{2}\ln T$ piece and the explicit prefactor T depend on temperature:

$$\begin{aligned} S = -\frac{\partial F}{\partial T} &= N k_B\Big[\ln \frac{V}{N\Lambda^3} + 1\Big] + N k_B T \cdot \frac{3}{2T} \\ &= N k_B\Big[\ln \frac{V}{N\Lambda^3} + \underbrace{1}_{\text{from } F} + \underbrace{\tfrac{3}{2}}_{\partial_T(\frac{3}{2}T\ln T)}\Big] = N k_B\Big[\ln \frac{V}{N\Lambda^3} + \frac{5}{2}\Big]. \end{aligned} \tag{28}$$

This is the *Sackur–Tetrode* equation. Its additive $\frac{5}{2}$ has a definite provenance: 1 from the indistinguishability remainder (the $+1$ in F, i.e. the $-\ln N!$ Stirling term) and $\frac{3}{2}$ from differentiating the thermal energy $\frac{3}{2}N k_B T$. Physically the two variable terms are the split we will track to the end: $\ln(V/N)$ is *configurational* (positional microstates—more room per particle, more states), and $-3\ln\Lambda = \frac{3}{2}\ln T + \text{const}$ is *thermal* (the momentum distribution broadens with T).

2.9 Intensive form, and the fate of the two $5/2$'s

Reduce (28) to a specific entropy. Per particle, in units of k_B,

$$\tilde{s} \equiv \frac{S}{N k_B} = \ln \frac{V}{N\Lambda^3} + \frac{5}{2}. \tag{29}$$

Use the ideal-gas law in the form $V/N = k_B T/p$ and $\Lambda^{-3} = (2\pi m k_B/h^2)^{3/2} T^{3/2}$:

$$\begin{aligned} \ln \frac{V}{N\Lambda^3} &= \ln \frac{k_B T}{p} + \frac{3}{2}\ln T + C_0 = \underbrace{\ln T}_{\text{from } V/N} - \ln p + \underbrace{\tfrac{3}{2}\ln T}_{\text{from } \Lambda^{-3}} + C_1 \\ &= \frac{5}{2}\ln T - \ln p + C_1, \qquad C_1 = \ln k_B + C_0. \end{aligned} \tag{30}$$

Substituting into (29),

$$\tilde{s} = \frac{5}{2}\ln T - \ln p + \Big(C_1 + \frac{5}{2}\Big) = \underbrace{\frac{5}{2}}_{=c_p/R}\ln T - \ln p + \text{const}. \tag{31}$$

There are *two* factors of $5/2$ here and they behave oppositely—this is the crux:

- The **additive** $\frac{5}{2}$ inherited from Sackur–Tetrode (29) is a pure constant. It is swallowed into the reference constant in (31) and cancels in every entropy *difference*. So yes—it drops out of anything physical; it survives only in the absolute, third-law-anchored entropy.
- The **coefficient** $\frac{5}{2}$ multiplying $\ln T$ is not a constant and does *not* drop out. Equation (30) shows it is assembled as $1 + \frac{3}{2}$: one power of $\ln T$ from the configurational $\ln(V/N) = \ln(k_B T/p)$, and $\frac{3}{2}$ from the thermal $\Lambda^{-3} \propto T^{3/2}$. For a monatomic gas this equals c_p/R.

Multiplying (31) by the specific gas constant $R = k_B/m$ (mass m per particle) turns $\tilde{s}$ into entropy per unit mass, and $\frac{5}{2}R = c_p$ for a monatomic gas, giving the building block

$$\boxed{s \;=\; c_p \ln T - R\ln p + \text{const.}} \tag{32}$$

2.10 Beyond monatomic: internal degrees of freedom

Sackur–Tetrode as derived assumes a structureless particle, so (28) does *not* apply verbatim to air (N_2, O_2) or water vapour (H_2O), which rotate and vibrate. Two things change, both controlled. First, the single-particle energy is a sum $\varepsilon = \varepsilon_{\mathrm{tr}} + \varepsilon_{\mathrm{int}}$ (translation plus rotation, vibration, electronic), so the partition function *factorises*,

$$z = z_{\mathrm{tr}}(T, V)\, z_{\mathrm{int}}(T), \tag{33}$$

with the internal factor a function of T alone (the internal level spacings do not depend on the box). Second, carrying z_{int} through (27)–(28) adds an internal energy $u_{\mathrm{int}} = k_B T^2\, \mathrm{d}\ln z_{\mathrm{int}}/\mathrm{d}T$ per particle and a matching internal entropy; the extra heat capacity is $c_{v,\mathrm{int}} = \mathrm{d}u_{\mathrm{int}}/\mathrm{d}T$.

The **equipartition theorem** makes this concrete: every quadratic degree of freedom that is thermally active contributes $\frac{1}{2}k_B$ to the per-particle heat capacity. Translation always gives $3 \times \frac{1}{2}$; rotation adds $2 \times \frac{1}{2}$ for a linear molecule (N_2, O_2, CO_2) or $3 \times \frac{1}{2}$ for a nonlinear one (H_2O); each active vibrational mode counts twice (kinetic + potential). Hence $c_v = \frac{f}{2}k_B$ over f active quadratic freedoms, $c_p = c_v + k_B$, and the monatomic ratio $c_p/R = \frac{5}{2}$ is replaced by $c_p/R = \frac{f}{2} + 1$.

Crucially, the *functional form* of the entropy is unchanged. Whenever c_p is constant—all active modes classical, none freezing in or out over the range of interest—integrating the thermodynamic identity $\mathrm{d}s = c_p\, \mathrm{d}T/T - R\, \mathrm{d}p/p$ (derived independently from the first and second laws in §3) gives

$$s = c_p \ln T - R \ln p + \mathrm{const} \tag{34}$$

for *any* ideal gas, with the material c_p. Sackur–Tetrode is just the monatomic special case that additionally fixes the absolute constant. For the atmosphere ($\approx$200–320 K) the diatomic rotational modes are fully excited while the stiff vibrational modes are largely frozen, so c_{pd} and c_{pv} (Table 1) are very nearly temperature-independent and (34) holds to high accuracy; the small residual $c_p(T)$ variation is absorbed into the effective constants and, for water, into the c_ℓ tuning of Remark 5.

Remark 4 (what the additive constant hides, and why it does not matter)**.** The reference constant in (32)/(34) absorbs Λ, the Sackur–Tetrode $\frac{5}{2}$, the internal-mode ground states, and each species' zero of energy. Because the ascent conserves a *difference* of entropies at fixed composition (§7) and buoyancy depends on temperature *differences* (§8), these constants never enter a physical prediction; the code is free to drop them, and does.

2.11 The bridge to $S(E)$: Laplace, Legendre, and why Z is the elegant object

The canonical machinery has now done real work—Z, differentiated twice, produced Sackur–Tetrode and the whole family (34)—so this is the right moment to step back and ask how it relates to the older, more geometric foundation: the microcanonical entropy $S(E)$, the logarithm of the phase-space volume at fixed energy. Making the link exact explains both why they agree and why the Z side carried every derivation above so much more lightly than the reservoir bookkeeping of §2.3 did.

The transform. Two objects must be kept apart. The *density of states*

$$\rho(E) = \mathrm{Tr}\ \delta(E - \hat{H}) = \sum_i \delta(E - E_i) \tag{35}$$

of §2.1—with $\delta(E - \hat{H}) = \sum_i \delta(E - E_i)\, |i\rangle\langle i|$ the operator function of $\hat{H}$ (evaluated on the spectrum and traced *after*, exactly as $Z = \mathrm{Tr}\, e^{-\beta \hat{H}} = \sum_i e^{-\beta E_i}$)—is a distribution, a comb of one spike per level; its logarithm is meaningless. The Boltzmann entropy instead uses the dimensionless *count*

$$\Omega(E) = \int_E^{E+\Delta E} \rho(E')\,\mathrm{d}E' \approx \bar{\rho}(E)\,\Delta E, \qquad S(E) = k_B \ln \Omega(E), \tag{36}$$

the number of levels in a macroscopically thin shell $[E, E+\Delta E]$, with $\bar{\rho}$ the smoothed envelope of the comb. Because the spectrum is spaced like e^{-N}, any such shell holds astronomically many levels, so Ω is a huge *smooth* number and $\ln \Omega$ is well defined and extensive; the window enters only through the subextensive $k_B \ln \Delta E$, negligible beside $S \sim k_B N$. (Ω and $\bar{\rho}$ thus agree *inside* the logarithm but are not the same object—they differ by the units-carrying ΔE, which is why ρ and Ω get different symbols.) Figure 4 draws all three objects at once for three oscillators. The partition function is the Laplace transform of the *density* ρ,

$$Z(\beta) = \int_0^\infty \mathrm{d}E\, \rho(E)\, e^{-\beta E} = \sum_i e^{-\beta E_i} \approx \int_0^\infty \mathrm{d}E\, \exp\Big[\tfrac{1}{k_B} S(E) - \beta E\Big], \tag{37}$$

and the middle equality is the crux: the exponential kernel integrates straight over the comb, so Z does *for free* the coarse-graining that $S(E)$ needed a window for—the "distribution → analytic function" elegance, met again below. It is invertible by the Bromwich integral $\rho(E) = \frac{1}{2\pi i} \int Z(\beta)\, e^{\beta E} \mathrm{d}\beta$: ρ and Z carry identical information. The oscillator shows this concretely—and more cleanly still, since its spectrum is genuinely discrete, so the "count" is an honest integer degeneracy needing no window at all. The microcanonical count $\Omega(M, n) = \binom{M+n-1}{n-1}$ of (14) is a genuine combinatorial sequence, but it is packaged whole by

$$Z(x) = \sum_{M \geq 0} \Omega(M, n)\, x^M = \sum_{M \geq 0} \binom{M+n-1}{n-1} x^M = \frac{1}{(1-x)^n}, \qquad x = e^{-\beta\hbar\omega}, \tag{38}$$

so Z is nothing but the *generating function* whose Taylor coefficients are the shell dimensions; a single $\Omega(M, n)$ comes back as the coefficient integral $\frac{1}{2\pi i} \oint Z(x)\, x^{-M-1} \mathrm{d}x$, whose saddle point gives the entropy asymptotics. "Laplace transform of the density of states" and "generating function of the counts" are one statement, continuous and discrete.

The thermodynamic limit is a Legendre transform. For a macroscopic system the integrand of (37) is sharply peaked, and Laplace's method evaluates it at the saddle $E^*(\beta)$ where the exponent is stationary,

$$\frac{\mathrm{d}}{\mathrm{d}E}\Big[\tfrac{1}{k_B} S(E) - \beta E\Big]_{E^*} = 0 \implies \frac{\mathrm{d}S}{\mathrm{d}E}\Big|_{E^*} = k_B \beta = \frac{1}{T}, \tag{39}$$

which is exactly the microcanonical definition of temperature. Back-substituting, $-k_B T \ln Z = E^* - T\, S(E^*) = F$: the free energy is the Legendre transform of $S(E)$, with β and E conjugate,

$$\ln Z(\beta) = \sup_E \Big[\tfrac{1}{k_B} S(E) - \beta E\Big], \qquad \tfrac{1}{k_B} S(E) = \inf_\beta \big[\ln Z(\beta) + \beta E\big], \tag{40}$$

with duality relations $\beta = \frac{1}{k_B} S'(E)$ and $E = -\partial_\beta \ln Z$. So the Legendre transform relating the ensembles (the table of §2.1) is just the $N \to \infty$ shadow of the exact Laplace transform (37): the saddle collapses the whole spectrum onto one energy E^* and one temperature. (This is the thermodynamic face of the Cramér/large-deviation duality between the scaled cumulant generating function $\ln Z$ and its rate function S/k_B.) Figure 5 draws both statements exactly for the oscillator: the tangent construction on the exact $S(E)$, and the saddle sharpening as $1/\sqrt{n}$.

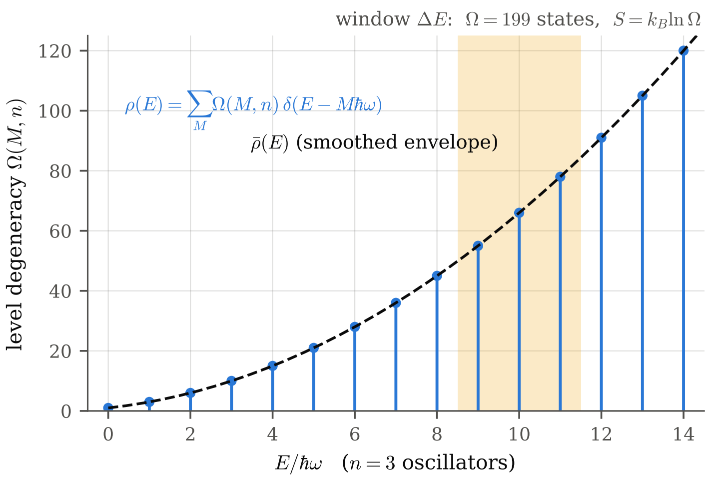

Figure 4: The two objects of (35)–(36), drawn for $n = 3$ oscillators. The density of states $\rho(E)$ is a comb of δ-spikes, one per level, with integer weights $\Omega(M,3) = \binom{M+2}{2}$ (stems); the smoothed envelope $\bar{\rho}(E)$ is the dashed curve; and a window ΔE (shaded) converts the comb into the dimensionless count $\Omega(E)$—here 199 states—whose logarithm is the Boltzmann entropy. The logarithm is taken of the count, never of the comb.

Why Z is the elegant object. Every awkward operation on $S(E)$ is an easy one on Z, because the Laplace transform converts them:

- **A distribution becomes an analytic function.** $\rho(E) = \operatorname{Tr}\delta(E - \hat{H})$ is a comb of spectral spikes, and $S(E)$ needs a coarse-graining window to define at all; $Z(\beta) = \operatorname{Tr} e^{-\beta\hat{H}}$ is a smooth, entire function of β for any spectrum bounded below—no delta functions, no window.

- **Convolution becomes multiplication—and the reservoir evaporates.** Independent systems ($\hat{H} = \hat{H}_1 + \hat{H}_2$) *convolve* their densities of states, $\rho = \rho_1 * \rho_2$, but simply *multiply* their partition functions, $Z = Z_1 Z_2$, so $\ln Z$ is manifestly extensive. The whole "system + reservoir, expand $\ln\Omega_{\text{res}}(E_{\text{tot}} - E_i)$" maneuver of §2.3 was a convolution evaluated by saddle point; in the Z picture one never introduces the reservoir—with the intensive β as primary, the Boltzmann weight $e^{-\beta E_i}/Z$ is immediate. The reservoir's only job was to fix $\beta = S'_{\text{res}}/k_B$, which Z takes as its argument.

- **$\ln Z$ generates the cumulants.** $\langle E\rangle = -\partial_\beta \ln Z$ and $\operatorname{Var}(E) = \partial_\beta^2 \ln Z = k_B T^2 C_V \geq 0$; every fluctuation is a derivative of one smooth function, and fluctuation–dissipation is one line. Energy fluctuations are not even representable on a single microcanonical E.

- **Z is what one computes.** Path integrals, transfer matrices, perturbation theory, and Monte Carlo all act on Z or $\ln Z$; counting states at fixed energy is the hard problem. The oscillator is the epitome: $\Omega(M,n) = \binom{M+n-1}{n-1}$ is a nontrivial count, while $Z = (1-x)^{-n}$ is one line (38).

When $S(E)$ is the truer object. The elegance has a precise price: the Legendre transform (40) can only return the *concave hull* of $S(E)$, so wherever $S(E)$ is non-concave the Z picture silently discards information—replacing the offending region by its Maxwell tangent (Fig. 6). That is exactly what happens at first-order phase transitions (the flat tangent is the latent heat), and in long-range or self-gravitating systems, at negative temperatures, and in small systems—all cases of genuine *ensemble inequivalence*, where the microcanonical $S(E)$ is faithful

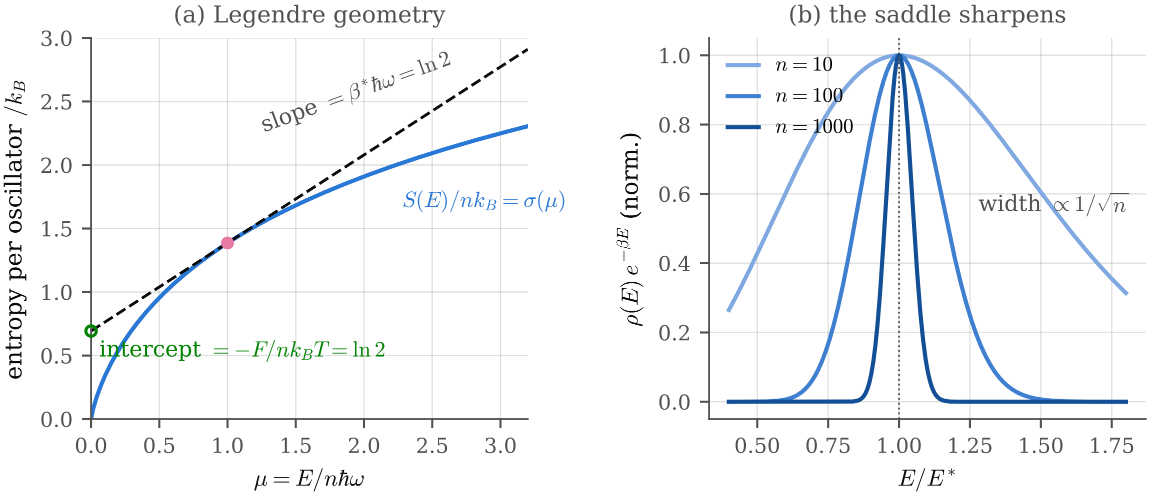

Figure 5: The bridge, drawn exactly for the oscillator. (a) The entropy per oscillator $\sigma(\mu)$ of (17) is concave; the tangent at $\mu^*=1$ has slope $\beta^*\hbar\omega=\ln 2$ (39) and intercept $-F/nk_BT=\ln 2$: the Legendre transform (40) *is* this tangent-and-intercept geometry. (b) The Laplace integrand $\rho(E)e^{-\beta E}$ of (37) at $\beta=\beta^*$, normalised to its peak: the saddle at E^* sharpens with relative width $1/\sqrt{n}$ (42)—the hard and soft constraints becoming operationally interchangeable.

and the canonical average is not. So the honest ranking is the geometric one: $S(E)$ is primary and information-complete; $Z(\beta)$ is its Laplace transform—smoother, factorising, fluctuation-generating, and matched to how a system is actually held (at fixed T, not fixed E)—and the two are fully equivalent precisely when $S(E)$ is concave, i.e. short-range interactions as $N\to\infty$. That is the parcel's regime, and the reason the intensive $s(T,p)$—not a fixed-energy count—is the right variable for the solver.

Hard constraint, soft constraint, and the separation of scales. The same transform settles what the microcanonical and canonical descriptions are to *each other*, and answers a natural worry: the closed whole is governed by exact energy conservation, yet a small piece of it obeys the Boltzmann distribution at a fixed temperature—these feel as if they should be the same statement, and are not quite. Fixing the total energy exactly is a *hard* constraint—a delta function $\delta(E_{\rm tot}-\hat H)$ projecting onto the energy shell, with *zero* energy fluctuation. Fixing only the *mean* energy through β is a *soft* constraint—the Boltzmann weight $e^{-\beta\hat H}$, with β a Lagrange multiplier (the Jaynes reading of §2.3) and the energy free to fluctuate. They are genuinely different ensembles, yet locked together by an identity: the hard projector is an integral of soft weights over imaginary temperature,

$$\delta(E_{\rm tot}-\hat H)=\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}{\rm d}\beta\; e^{\beta(E_{\rm tot}-\hat H)},\tag{41}$$

which is just the inverse Bromwich transform of (37) read as an operator identity—the sharp shell is a coherent superposition of Boltzmann weights at *every* β. “Hard” and “soft” are one function and its Laplace transform, not two physical laws. What makes them *equivalent* for a macroscopic system is a separation of scales, entering in two places.

Whole-system equivalence, guarded by $1/\sqrt{N}$. For large N the β-integral (41)—equivalently the Laplace integral (37)—is dominated by a single real saddle β^* where $S'(E_{\rm tot})=1/T$, so the sharp constraint collapses onto *one* effective temperature. The width of that collapse is the energy fluctuation the soft constraint permits, which by the cumulant identity ${\rm Var}(E)=$

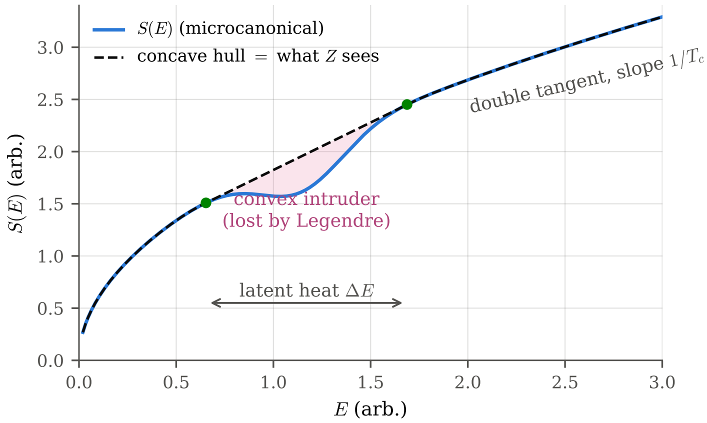

Figure 6: Where the elegance has a price (schematic). A non-concave $S(E)$ carries a convex intruder (shaded); the Legendre transform (40) can return only the concave hull (dashed), whose double tangent of slope $1/T_c$ replaces the intruder entirely. The canonical picture is blind to the shaded region—at T_c it jumps across the interval ΔE (the latent heat) rather than passing through it. This is ensemble inequivalence: first-order transitions, long-range/self-gravitating systems, negative temperatures, small systems. In the parcel's regime (short-range, $N \to \infty$) $S(E)$ is concave and nothing is lost.

$\partial_\beta^2 \ln Z = k_B T^2 C_V$ above and the extensivity $C_V, \langle E \rangle \sim N$ is

$$\frac{\sqrt{\mathrm{Var}(E)}}{\langle E \rangle} = \frac{\sqrt{k_B T^2 C_V}}{\langle E \rangle} \sim \frac{\sqrt{N}}{N} = \frac{1}{\sqrt{N}}. \tag{42}$$

Energy fluctuations are $O(\sqrt{N})$ absolutely but $O(1/\sqrt{N})$ relatively: the soft constraint is *effectively* as sharp as the hard one, and the two ensembles agree on every intensive quantity to that order.

Subsystem equivalence, guarded by N_S/N_B. The framing closest to the parcel is a small subsystem S inside a hard-constrained whole $S + B$. Marginalising the shell over the bath gives the subsystem's energy distribution

$$p(E_S) \;\propto\; \Omega_S(E_S)\,\Omega_B(E_{\mathrm{tot}} - E_S) \;=\; \Omega_S(E_S)\, e^{S_B(E_{\mathrm{tot}} - E_S)/k_B}, \tag{43}$$

which is exactly $p_i \propto \Omega_{\mathrm{res}}(E_{\mathrm{tot}} - E_i)$ of (6). Because $E_S \ll E_{\mathrm{tot}}$—that is, $N_S \ll N_B$—the bath entropy is linear over the range E_S explores, $S_B(E_{\mathrm{tot}} - E_S) \simeq S_B(E_{\mathrm{tot}}) - E_S/T$, and (43) becomes the Boltzmann weight $p(E_S) \propto \Omega_S(E_S)\, e^{-\beta E_S}$: **the hard constraint on the whole is a soft constraint on the part.** The bath's temperature is *stiff*—$\mathrm{d}\beta/\mathrm{d}E = -1/(k_B T^2 C_B)$ with $C_B \sim N_B$—so draining E_S barely moves β; the discarded curvature is the finite-bath back-reaction, of order $E_S/E_{\mathrm{tot}} \sim N_S/N_B$. This linearisation *is* the Taylor truncation (7) performed by hand in §2.3, now with its small parameter named. Figure 7 shows the whole statement in one frame: the exact hard-constraint marginal at $n = 3$ is visibly not exponential (its support even ends at $\varepsilon = E$, the whole-bath back-reaction at full strength), and flows into the Boltzmann weight as n grows.

When the guard fails. Both equivalences are asymptotic and conditional. If $S(E)$ is non-concave the saddle of (41) is not unique—two compete, at a first-order transition—and the ensembles genuinely diverge, the very ensemble inequivalence just discussed; if the system is small, (42) is not negligible and the soft constraint is visibly looser than the hard one. So the instinct that the two "ought to be the same thing" is right only in the limit: they are one Laplace pair, made *operationally* interchangeable by the $1/\sqrt{N}$ and N_S/N_B separations, and distinguishable again the moment those separations close.

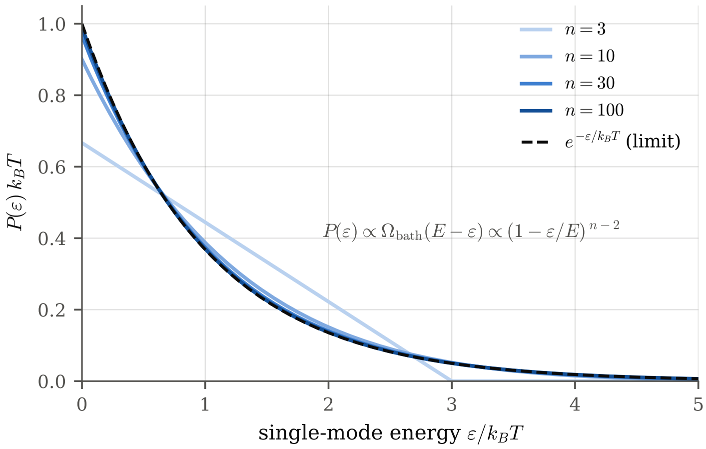

Figure 7: The hard constraint on the whole becoming a soft constraint on the part. Exact marginal $P(\varepsilon) \propto \Omega_{\text{bath}}(E - \varepsilon) = (1 - \varepsilon/E)^{n-2}(n-1)/E$ of one classical oscillator in an isolated n-oscillator system at fixed total energy $E = nk_BT$ (43). At $n = 3$ the finite bath is obvious—the distribution is not exponential and its support ends at $\varepsilon = E$; by $n = 100$ it lies on the Boltzmann limit $e^{-\varepsilon/k_BT}$ (dashed). The convergence rate is the N_S/N_B guard of the text.

Having set Z against its microcanonical shadow, we can cash the whole construction out for the parcel: the building block below.

2.12 The building block, and how it feeds everything below

Equation (34) is the payoff of this section and the answer to "how does the ideal-gas entropy relate to everything that follows". It is not a monatomic curiosity: it is the single functional shape that each gaseous constituent of the parcel instantiates *once*. Dry air and water vapour are both ideal gases, so each contributes

$$s_k = c_{p,k} \ln T - R_k \ln(\text{its own partial pressure}) + \text{const}, \tag{44}$$

differing only in which c_p, which R, and—by Dalton's law (§4)—which partial pressure it sees. The total specific entropy of the parcel is then the mixing-ratio-weighted sum of these identical blocks plus a condensed-phase term (Gibbs additivity, §4); once phase equilibrium is used to reintroduce the latent heat (§5), that sum is exactly Emanuel's (60). In short: derive the block once here, and "everything below" is bookkeeping—which partial pressure, which c_p, which mixing ratio—layered on top of (34).

3 Classical scaffolding: why s is a state function

Statistical mechanics gives s; classical thermodynamics tells us it is a *function of state*—indispensable, since the solver evaluates s at a point (T, p, r_v) with no reference to how the parcel got there.

3.1 First and second laws

The first law for a closed system is $\mathrm{d}U = \delta Q - \delta W$, with $\delta W = p\,\mathrm{d}V$ for quasi-static volume work. Neither δQ nor δW is an exact differential: $\oint \delta Q \neq 0$. The second law supplies an *integrating factor*. In Carathéodory's formulation, the empirical fact that arbitrarily close to any equilibrium state there exist states not reachable by any adiabatic path ($\delta Q = 0$) is equivalent, for the Pfaffian form δQ, to the existence of an integrating denominator; that denominator is the absolute temperature T, and

$$\mathrm{d}s \;=\; \frac{\delta Q_{\mathrm{rev}}}{T} \tag{45}$$

is an *exact* differential. Its integral, s, is a state function. This is the precise sense in which "entropy exists".

3.2 Recovering $s = c_p \ln T - R \ln p$ thermodynamically

For an ideal gas, $pV = nRT$ and $U = U(T)$ with $\mathrm{d}U = c_v\,\mathrm{d}T$ (Joule's law). Then $\delta Q_{\mathrm{rev}} = \mathrm{d}U + p\,\mathrm{d}V = c_v\,\mathrm{d}T + p\,\mathrm{d}V$, and with $p/T = R/v$ (specific volume v),

$$\mathrm{d}s = \frac{c_v\,\mathrm{d}T}{T} + \frac{R\,\mathrm{d}v}{v} = \frac{c_p\,\mathrm{d}T}{T} - \frac{R\,\mathrm{d}p}{p}, \tag{46}$$

using Mayer's relation $c_p - c_v = R$ and $\mathrm{d}(\ln pv) = \mathrm{d}\ln(RT)$ to switch variables. Integrating returns (32), now with the material c_p. Statistical mechanics and classical thermodynamics agree, as they must; the exactness of $\mathrm{d}s$ (equivalently the Maxwell relation $\partial^2 s/\partial T\partial p = \partial^2 s/\partial p\partial T$) is what lets `entropy_S` be a pure function of its three arguments.

4 The parcel as a multicomponent mixture

4.1 Composition and the per-dry-air convention

The parcel is dry air plus water in vapour and (once saturated) condensed form. Masses are bookkept as *mixing ratios* relative to the dry-air mass m_d:

$$r_v = \frac{m_{\mathrm{vapour}}}{m_d}, \qquad r_\ell = \frac{m_{\mathrm{liquid}}}{m_d}, \qquad r_t = r_v + r_\ell. \tag{47}$$

Entropy is quoted *per unit mass of dry air* because m_d is the one mass that is conserved along the ascent (water changes phase, but dry air is inert and neither created nor removed); dividing by it gives a bookkeeping base that does not move. This is why (1) carries R_d bare but weights the water terms by mixing ratios.

4.2 Dalton's law and the vapour–pressure relation

Each gas in an ideal mixture behaves as if it alone occupied the volume, so the total pressure is the sum of partial pressures (Dalton): $p = p_d + e$, with e the vapour partial pressure and $p_d = p - e$ the dry-air partial pressure. The mixing ratio and the partial pressure are two encodings of the same amount of vapour; equating $r_v = m_v/m_d = (\varepsilon^{-1}e)/(p-e)$ with $\varepsilon = R_d/R_v$ gives the mutually inverse maps

$$e = \frac{r_v\,p}{\varepsilon + r_v}, \qquad r_v = \frac{\varepsilon\,e}{p - e}, \tag{48}$$

which are exactly `utilities.ev` and `utilities.rv`. The appearance of $p - e$ (not p) inside the dry-air logarithm of (1) is Dalton's law at work: the dry air sees only its own partial pressure.

4.3 Additivity of entropy: Gibbs' theorem

For an ideal-gas mixture the entropy is the sum of the entropies each component would have alone at its own partial pressure and the common temperature (Gibbs' theorem—there is no entropy of mixing beyond what the partial pressures already encode, because ideal gases do not interact). Hence we may build s additively:

$$s \;=\; s_d \;+\; r_v\, s_v \;+\; r_\ell\, \sigma_\ell, \tag{49}$$

s_d the specific entropy of dry air, s_v that of vapour, and σ_ℓ that of liquid water, each per unit mass of its own species. We now need the three pieces.

5 Water substance and phase equilibrium

The gaseous pieces use (32):

$$s_d = c_{pd} \ln T - R_d \ln p_d + \mathrm{const}_d, \qquad s_v = c_{pv} \ln T - R_v \ln e + \mathrm{const}_v. \tag{50}$$

Liquid water is nearly incompressible, so its entropy is essentially thermal,

$$\sigma_\ell = c_\ell \ln T + \mathrm{const}_\ell. \tag{51}$$

What ties the vapour and liquid constants together—and produces the latent term in (1)—is phase equilibrium.

5.1 Chemical potential and the Clausius–Clapeyron relation

At a flat interface, vapour and liquid coexist when their specific Gibbs free energies (chemical potentials per unit mass) are equal, $g_v(T, e_s) = g_\ell(T)$, which *defines* the saturation vapour pressure $e_s(T)$. Differentiating along the coexistence curve with $\mathrm{d}g = -s\,\mathrm{d}T + v\,\mathrm{d}p$ gives

$$-s_v\,\mathrm{d}T + v_v\,\mathrm{d}e_s \;=\; -\sigma_\ell\,\mathrm{d}T + v_\ell\,\mathrm{d}e_s \;\Longrightarrow\; \frac{\mathrm{d}e_s}{\mathrm{d}T} = \frac{s_v - \sigma_\ell}{v_v - v_\ell}. \tag{52}$$

The reversible latent heat is the enthalpy of the isothermal, isobaric phase change, $L_v = T(s_v - \sigma_\ell)$, i.e. the *entropy of vaporisation* is

$$s_v - \sigma_\ell = \frac{L_v}{T}. \tag{53}$$

With $v_v \gg v_\ell$ and $v_v = R_v T/e_s$ (ideal vapour), this is the Clausius–Clapeyron equation

$$\frac{\mathrm{d}e_s}{\mathrm{d}T} = \frac{L_v\, e_s}{R_v\, T^2}. \tag{54}$$

5.2 Kirchhoff's relation and the coded fits

The latent heat itself depends on temperature. Applying $\mathrm{d}L_v/\mathrm{d}T = c_{pv} - c_\ell$ (Kirchhoff's law: the two phases warm at different specific heats) and integrating from 0°C,

$$L_v(T) = L_v^0 + (c_{pv} - c_\ell)\,(T - T_0), \tag{55}$$

which is exactly `utilities.Lv` (Table 1: $L_v^0 = 2.501 \times 10^6$, slope $c_{pv} - c_\ell = -630$). Substituting (55) into (54) and integrating yields a closed expression for $e_s(T)$; the code uses the algebraically cheaper August–Roche–Magnus fit

$$e_s(T) = 6.112 \,\exp\!\Big(\frac{17.67\, T_C}{243.5 + T_C}\Big)\ \mathrm{hPa}, \qquad T_C = T - 273.15, \tag{56}$$

i.e. `utilities.es_cc`, which reproduces the Clausius–Clapeyron integral to sub-percent accuracy over atmospheric temperatures. Equations (53)–(56) are all we need from phase physics.

Remark 5 (the "modified" $c_\ell = 2500\,\mathrm{J\,kg^{-1}\,K^{-1}}$)**.** The true specific heat of liquid water is $\approx 4190\,\mathrm{J\,kg^{-1}\,K^{-1}}$; `tcpyPI` (following `pcmin`) uses 2500. This is a deliberate effective value that folds the ice phase and the temperature dependence of L_v into a single linear Kirchhoff slope $c_{pv} - c_\ell = -630$, tuned so that (55) tracks the observed $L_v(T)$ across the mixed-phase range the parcel traverses. It changes the reference thermodynamics slightly but not the structure of any derivation here.

6 Assembling the total specific entropy

Insert (50)–(51) into the additive decomposition (49) and use $r_\ell = r_t - r_v$:

$$\begin{aligned} s &= \left[c_{pd}\ln T - R_d \ln p_d\right] + r_v\left[c_{pv}\ln T - R_v \ln e\right] + (r_t - r_v)\, c_\ell \ln T + \text{const} \\ &= c_{pd}\ln T - R_d \ln p_d + r_t\, c_\ell \ln T + r_v\left[(c_{pv} - c_\ell)\ln T - R_v \ln e\right] + \text{const}. \end{aligned} \tag{57}$$

The bracket on the vapour term is where the latent heat re-enters. Evaluate (53) using (50)–(51) *at saturation* ($e = e_s$):

$$\frac{L_v}{T} = s_v - \sigma_\ell = (c_{pv} - c_\ell)\ln T - R_v \ln e_s + (\text{const}_v - \text{const}_\ell). \tag{58}$$

Solve (58) for $(c_{pv} - c_\ell)\ln T$ and substitute into the bracket of (57):

$$\begin{aligned} (c_{pv} - c_\ell)\ln T - R_v \ln e &= \left[\tfrac{L_v}{T} + R_v \ln e_s - (\text{const}_v - \text{const}_\ell)\right] - R_v \ln e \\ &= \frac{L_v}{T} - R_v \ln\frac{e}{e_s} - (\text{const}_v - \text{const}_\ell) = \frac{L_v}{T} - R_v \ln H + \text{const}. \end{aligned} \tag{59}$$

Putting this back into (57) and absorbing all reference constants,

$$\boxed{s = (c_{pd} + r_t\, c_\ell)\ln T - R_d \ln(p - e) + \frac{L_v\, r_v}{T} - r_v\, R_v \ln H.} \tag{60}$$

This is Emanuel (1994), eq. 4.5.9, and term-for-term it is (1) and `utilities.entropy_S`. The four terms carry the four physical meanings we tracked from the start:

Term	Physical content
$(c_{pd} + r_t c_\ell)\ln T$	thermal entropy of dry air + all condensate carried
$-R_d \ln(p - e)$	configurational entropy of the dry air (Dalton partial pressure)
$+L_v r_v/T$	latent entropy stored in the vapour fraction (Clausius–Clapeyron)
$-r_v R_v \ln H$	dilution entropy of sub-saturated vapour ($H < 1 \Rightarrow\, >0$)

Remark 6 (the single-argument `R`, and reduction below saturation)**.** The code's (1) uses one mixing ratio `R` in every water term, i.e. it identifies $r_v = r_t = R$. That is exact wherever the parcel is unsaturated, which is precisely where `entropy_S` is called: at the launch state, to set the conserved reference s_0. Below the lifting condensation level all water is vapour, $r_\ell = 0$, and (60) is the familiar dry/unsaturated entropy with $H < 1$. The generalisation to $r_v < r_t$ (condensate present) is carried implicitly by the saturated form used in the inversion, (69) below, where $H = 1$ retires the last term.

7 Why s is conserved on the ascent

7.1 The exact entropy budget

For the parcel as a closed system, the Clausius inequality splits any change of entropy into a reversible flux and an irreversible production:

$$\mathrm{d}s = \underbrace{\frac{\delta Q}{T}}_{\text{flux}} + \underbrace{\mathrm{d}s_{\text{irr}}}_{\geq 0 \text{ (production)}} . \tag{61}$$

The parcel model annihilates both terms by two idealisations.

Adiabatic: no flux, $\delta Q = 0$. Convective ascent is fast compared with the time for heat to diffuse or radiate across the parcel boundary. Formally, the ratio of the advective timescale H_ρ/w (density scale height over vertical velocity, minutes) to the thermal-diffusion and radiative timescales (hours to days) is $\ll 1$, so $\delta Q \approx 0$ over the ascent. The first term of (61) vanishes.

Reversible: no production, $\mathrm{d}s_{\text{irr}} = 0$. The parcel is assumed to remain in internal thermodynamic equilibrium—uniform T and p, no turbulent entrainment or mixing with the environment, and, crucially, phase change occurring exactly at saturation. At saturation the chemical potentials are equal, $g_v = g_\ell$ (§5), so transferring water across the phase boundary produces *no* entropy: condensation onto a parcel held at $H = 1$ is reversible. (Condensing sub- or super-saturated vapour would carry a finite Δg per unit mass and generate $\mathrm{d}s_{\text{irr}} > 0$.) The second term of (61) vanishes.

Together, $\delta Q = 0$ and $\mathrm{d}s_{\text{irr}} = 0$ give

$$\boxed{\mathrm{d}s = 0 \quad \text{along the ascent}} \tag{62}$$

—the ascent is *isentropic*. This is the entire justification for holding `entropy_S` fixed.

7.2 Closed for water: reversible vs. pseudo-adiabatic

Conservation of (60) as a function of (T, p) alone additionally requires that the water inventory not change: the "reversible" adiabat keeps all condensate suspended in the parcel, so $r_t = \text{const}$ and the parcel is a closed system for water. In the code this is the default `ascent_flag`$= 0$; the density temperature (§8) then carries the full water loading r_t. The alternative `ascent_flag`$= 1$ (*pseudo*-adiabatic) removes condensate the instant it forms: the system is then *open*, r_t is not conserved, and the invariant is not (60) but an approximate pseudo-entropy. The distinction is physical, not numerical.

7.3 The microscopic reading

Statistical mechanics gives (62) a clean interpretation. A reversible adiabatic change is a *slow* deformation of the parcel's Hamiltonian (volume and level spacings change as p and T change) with no coupling to a heat bath. Two theorems then apply. By **Liouville's theorem**, phase-space volume is invariant under Hamiltonian flow; by the **adiabatic theorem**, a sufficiently slow deformation preserves the occupation probability of each state (no transitions between energy shells). The accessible phase-space volume Ω is therefore unchanged, and by (3) so is S; equivalently every p_i in the Gibbs form (4) is frozen, so $-k_B \sum p_i \ln p_i$ is constant even as the individual energy levels move. Isentropic literally means *the microstate count does not change—the levels merely shift.* Note that the two senses of "adiabatic" (thermodynamic: no heat; mechanical: slow) here coincide, which is why the slow, heat-free ascent lands on a single adiabat. This is the closed-system twin of §2.4: there the same no-reservoir, single-Hamiltonian stance fixes *which* weights p_i are canonical (a pure state's reduced density matrix); here it explains why those weights are *frozen* under the reversible ascent. Together they carry both halves of the Gibbs entropy (4)—the value of the weights and their constancy—without ever invoking a heat bath.

8 The moist adiabat and buoyancy

8.1 From $\mathrm{d}s = 0$ to a path

Because s is a state function, (62) is a differential constraint that pins the path in the (T, p) plane:

$$0 = \mathrm{d}s = \frac{\partial s}{\partial T}\,\mathrm{d}T + \frac{\partial s}{\partial p}\,\mathrm{d}p \implies \frac{\mathrm{d}T}{\mathrm{d}p} = -\frac{\partial s/\partial p}{\partial s/\partial T}. \tag{63}$$

This ODE is the moist adiabat. Two regimes:

Unsaturated (below the LCL): the dry adiabat. With $r_v = r_t$ fixed and no condensation, the dominant balance of (60) is $s \approx (c_{pd} + r_t c_\ell)\ln T - R_d \ln p_d + \text{const}$. Holding s and the water constant and treating the small vapour corrections as constant, $\mathrm{d}s = 0$ gives $c_p\,\mathrm{d}\ln T = R_d\,\mathrm{d}\ln p$ (with the appropriate moist c_p), i.e. Poisson's relation

$$T \propto p^{R_d/c_p}, \qquad \theta \equiv T\Big(\frac{p_0}{p}\Big)^{R_d/c_p} = \text{const}, \tag{64}$$

the conserved (virtual) potential temperature. This is exactly the branch the CAPE routine takes below the condensation level, where it writes $T_G = T_P\,(p/p_P)^{R_d/c_{pd}}$.

The lifting condensation level. As the unsaturated parcel cools along (64), $e_s(T)$ falls (Clausius–Clapeyron) while e falls only with pressure; $H = e/e_s$ rises to 1. The pressure at which $H = 1$ is the *lifting condensation level* (LCL), `utilities.e_pLCL` in the code. Above it the parcel is saturated.

Saturated (above the LCL): the moist adiabat proper. Now $r_v = r_s(T, p) = \varepsilon e_s(T)/(p - e_s)$ is pinned to saturation and $r_\ell = r_t - r_s$ is condensate. Substituting into (63) (equivalently differentiating (60) at $H = 1$) gives the saturated adiabatic lapse rate; its steepness is set by the latent-heat release, and the temperature sensitivity is dominated by $\mathrm{d}r_s/\mathrm{d}T$ through Clausius–Clapeyron. Rather than integrate this ODE, the solver conserves s algebraically—the inversion of §9.

8.2 Density temperature and buoyancy

The dynamically relevant temperature is the *density temperature* T_ρ, defined so that $p = \rho\,R_d\,T_\rho$ holds with the dry-air gas constant despite the moisture. Accounting for vapour (which lightens the air, $R_v > R_d$) and suspended condensate (which weighs it down),

$$T_\rho = T\,\frac{1 + r_v/\varepsilon}{1 + r_t}, \tag{65}$$

exactly `utilities.Trho(T,` r_t`,` r_v`)`. The buoyancy (per unit mass) that drives the parcel is the fractional density deficit at equal pressure, i.e. the density-temperature excess over the environment,

$$b \;=\; g\,\frac{T_\rho^{\mathrm{p}} - T_\rho^{\mathrm{e}}}{T_\rho^{\mathrm{e}}}, \tag{66}$$

and hydrostatic balance for the environment, $\mathrm{d}z = -(R_d T_\rho^{\mathrm{e}}/g)\,\mathrm{d}\ln p$, turns its vertical integral into the pressure integral

$$\mathrm{CAPE} = \int b\,\mathrm{d}z = \int_p^{p_{\mathrm{launch}}} R_d\,(T_\rho^{\mathrm{p}} - T_\rho^{\mathrm{e}})\,\mathrm{d}\ln p, \tag{67}$$

exactly the $R_d\,\Delta T_\rho\,\Delta\ln p$ increments the CAPE routine accumulates level by level. The parcel's self-determined adiabat (§7) is compared against the sounding; the environment enters only here, in the comparison, and the parcel's own $T(p)$ came entirely from conserving s.

9 The inversion: recovering T from s

9.1 Definition and well-posedness

At each new pressure level the solver knows the conserved entropy s_0 and must find the temperature consistent with it. This is a function *inversion*: given the forward map $T \mapsto s(T,p)$ at fixed p (and r_t), solve

$$s(T,p) = s_0 \qquad \text{for } T = s^{-1}(s_0;p). \tag{68}$$

An inverse exists and is unique wherever the forward map is strictly monotone. Differentiate the saturated entropy—the relevant branch for the inversion,

$$s_{\mathrm{sat}}(T) = (c_{pd} + r_t\,c_\ell)\ln T - R_d\ln(p - e_s) + \frac{L_v\,r_s}{T}, \tag{69}$$

(the $-r_v R_v \ln H$ term is gone because $H=1$)—with respect to T at fixed p. Using Clausius–Clapeyron (54) for $\mathrm{d}r_s/\mathrm{d}T$, the dominant terms give

$$\frac{\partial s_{\mathrm{sat}}}{\partial T} = \frac{1}{T}\left(c_{pd} + r_t\,c_\ell + \frac{L_v^2\,r_s}{R_v\,T^2}\right) \;>\; 0. \tag{70}$$

Every term is manifestly positive, so $s_{\mathrm{sat}}(\cdot)$ is strictly increasing and (68) has a unique root. By the inverse function theorem the inverse is itself C^1 with derivative $(s^{-1})' = 1/(\partial s/\partial T)$. The inversion is well posed—this is the "smooth, strictly monotone entropy relation" invoked without proof in the companion analysis. Figure 8(a) draws (69) with the code's own formulas: at each pressure the curve is strictly increasing, the level line $s = s_0$ cuts it exactly once, and the family of intersections *is* the moist adiabat of §8.

9.2 Newton iteration, and its appearance in the code

There is no closed form for s^{-1}, so the solver inverts numerically by Newton–Raphson on the residual $F(T) = s_{\mathrm{sat}}(T) - s_0$:

$$T_{n+1} = T_n - \frac{F(T_n)}{F'(T_n)} = T_n + \frac{s_0 - s_{\mathrm{sat}}(T_n)}{\partial s_{\mathrm{sat}}/\partial T}. \tag{71}$$

This is exactly `solve_temperature_from_entropy`: its variable `SG` is $s_{\mathrm{sat}}(T_n)$ of (69), its `SL` is the derivative (70),

$$\mathtt{SL} = \frac{c_{pd} + r_t\,c_\ell + L_v^2 r_s/(R_v T^2)}{T},$$

and the update `TGNEW = TG + AP*(S-SG)/SL` is (71) with a relaxation factor `AP` (smaller for the first steps, then unity) guarding against overshoot from a poor initial guess. Quadratic convergence follows from $F \in C^2$ and $F' \neq 0$, both guaranteed by (70). The positivity of `SL` is not a numerical convenience; it is the inverse-function-theorem hypothesis, and (70) shows it holds term by term.

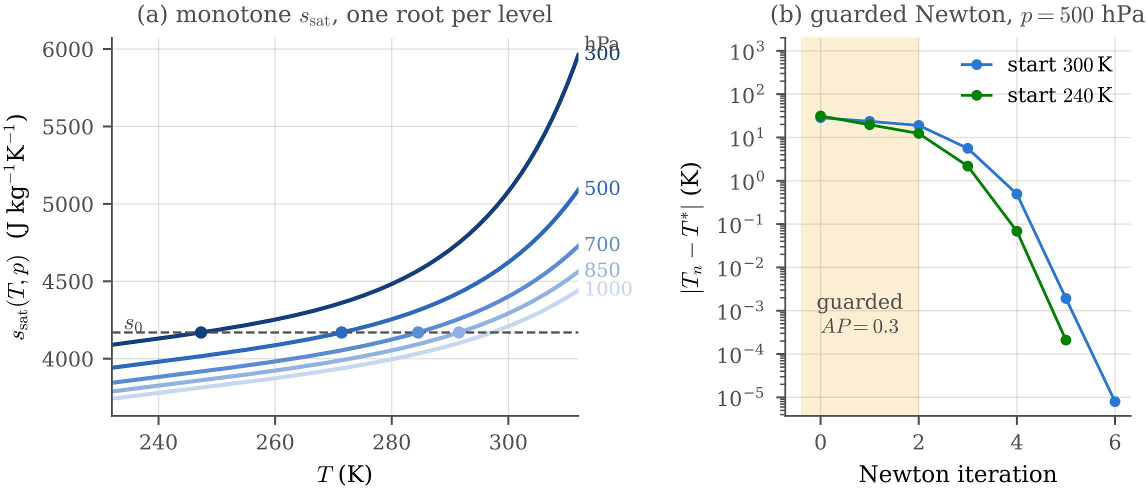

Figure 8: The inversion, computed with the code's own algebra (the script transcribes `es_cc`, `Lv`, `ev/rv`, `entropy_S` and the solver loop, and cross-checks each against the source docstring examples). (a) Saturated entropy (69) against temperature for five pressure levels, with the conserved $s_0 =$ `entropy_S`(300 K, 0.018, 1000 hPa) as the dashed level line; by the monotonicity (70) each curve crosses it exactly once (dots; the 1000 hPa level sits below the LCL of this parcel, so its crossing is not marked), and the dots trace the moist adiabat $T(p)$. (b) Error of the guarded Newton iteration (71) at 500 hPa from a warm and a cold start: the first two steps are relaxed (`AP` $= 0.3$, shaded), after which the error collapses quadratically to the 10^{-3} K stopping tolerance.

Remark 7 (why this is the "entropy relation", not a lapse-rate integral)**.** One could instead integrate the moist-adiabat ODE (63) in p. Conserving s algebraically via (68) is preferable: it carries no accumulated integration error, it makes the conserved quantity exact at every level by construction, and it exposes the monotonicity (70) that guarantees a unique parcel temperature. The price—a root-find per level—is cheap and quadratically convergent.

10 Synthesis

The chain is now complete and first-principled at every link:

1. Entropy is a microstate count, $S = k_B \ln \Omega = -k_B \sum p_i \ln p_i$ (§2); for an ideal gas it evaluates to $s = c_p \ln T - R \ln p + \text{const}$ (Sackur–Tetrode).

2. The Boltzmann weights need no external reservoir: canonical typicality derives them from a single pure state of one closed system—exactly, for n oscillators—with the equal-*a-priori* postulate demoted to a concentration-of-measure theorem, its exceptional states identified, and the separate (dynamical) status of ergodicity flagged (§2.4). Nor was the flat prior itself load-bearing: any weighting whose bath-conditioned tilt is $\ll \beta$ gives the same physics, sorting all priors into a four-tier gauge hierarchy whose only dangerous member is a tilt conjugate to an observable in play (§2.5).

3. The microcanonical $S(E)$ and canonical $Z(\beta)$ are Laplace duals whose thermodynamic-limit shadow is the Legendre transform; hard (fixed-E) and soft (fixed-β) constraints are operationally interchangeable under the $1/\sqrt{N}$ and N_S/N_B separations of scale, and Z is the smoother, factorising, cumulant-generating side of the pair wherever $S(E)$ is concave (§2.11).

4. The second law makes s an exact state function via the integrating factor $1/T$ (§3), so it may be evaluated pointwise.

5. The parcel is a Dalton mixture; entropy adds over components at their partial pressures (§4), and phase equilibrium ($g_v = g_\ell$) yields Clausius–Clapeyron and the vaporisation entropy L_v/T (§5).

6. Summing dry air, vapour, and condensate gives Emanuel's total specific entropy (60)—`utilities.entropy_S` exactly (§6).

7. It is conserved because the ascent is adiabatic (no flux) and reversible (no production, phase change at saturation), and reversible ascent keeps r_t fixed (§7); microscopically, Liouville plus the adiabatic theorem freeze Ω.

8. Conserving s defines the moist adiabat (63) and, via the strictly monotone (70), a well-posed inversion $s(T, p) = s_0 \mapsto T$ that `solve_temperature_from_entropy` performs by guarded Newton iteration (§8–9).

Every constant, fit, and function named above appears in the `tcpyPI` source—in `constants.py`, `utilities.py`, and `pi.py`; the physics that justifies each is derived here rather than asserted, and every figure is regenerated from those same formulas by `scripts/foundations_figs.py` (Fig. 8 by running the solver's own transcribed algebra, cross-checked against the source docstring examples). The one modelling choice worth flagging for a physicist reading the code is the effective $c_\ell = 2500\,\mathrm{J\,kg^{-1}\,K^{-1}}$ (Remark 5); everything else is textbook thermodynamics assembled in the meteorologist's per-dry-air bookkeeping.